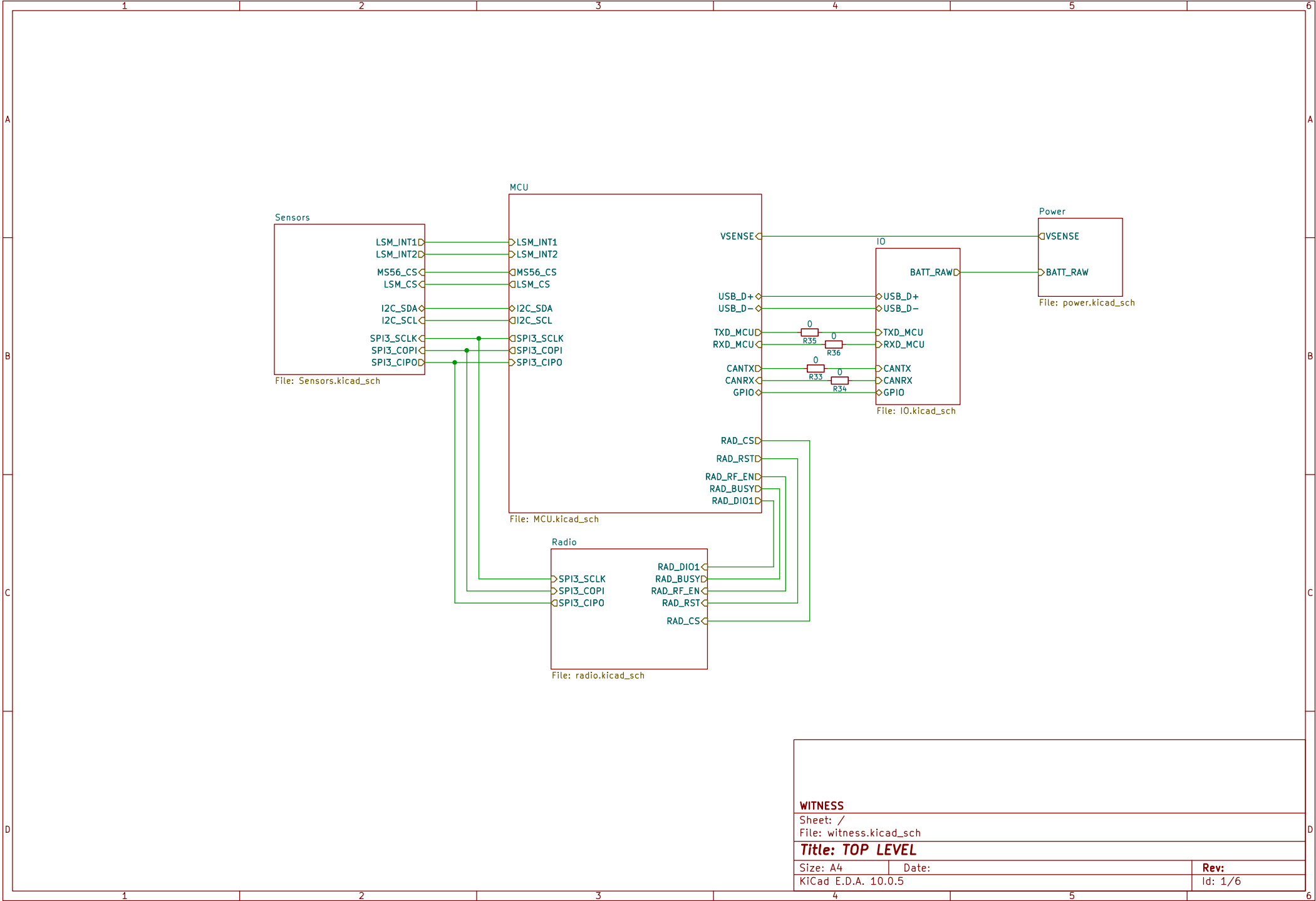




WITNESS

Sheet: /
File: witness.kicad_sch

Title: TOP LEVEL

Size: A4
KiCad E.D.A. 10.0.5

Date:

Rev:
Id: 1/6

Table 3-1. Default Configuration of Strapping Pins

Strapping Pin	Default Configuration	Bit Value
GPIO0	Weak pull-up	1
GPIO3	Floating	—
GPIO45	Weak pull-down	0
GPIO46	Weak pull-down	0

3.1 Chip Boot Mode Control

GPIO0 and GPIO46 control the boot mode after the reset is released. See Table 3-3 Chip Boot Mode Control.

Table 3-3. Chip Boot Mode Control

Boot Mode	GPIO0	GPIO46
SPI boot mode	1	Any value
Joint download boot mode ¹	0	0

¹ Bold marks the default value and configuration.

² Joint Download Boot mode supports the following download methods:

- USB Download Boot:
 - ~ USB-Serial-JTAG Download Boot
 - ~ USB-OTIO Download Boot
- UART Download Boot

Figure 1-1. ESP32-S3 Series Nomenclature

Table 1-1. ESP32-S3 Series Comparison

Part Number ¹	In-Package Flash ²	In-Package PSRAM	Ambient Temp. ³	VDD_SPI Voltage ⁴	Chip Version
ESP32-S3	—	—	-40 ~ 105 °C	3.3 V/1.8 V	v0.1/v0.2
ESP32-S3FN8	8 MB (Quad SPI) ⁵	—	-40 ~ 85 °C	3.3 V	v0.1/v0.2

The VDD_SPI voltage can be:

- (Default) 3.3 V supplied by VDD0P9_RTC via R_{PU}
- 1.8V supplied by the Flash Voltage Regulator

The voltage is determined by VDD0P9_RTC, VDD0P9_RTC, and VDD0P9_RTC.

Table 3-4. VDD_SPI Voltage Control

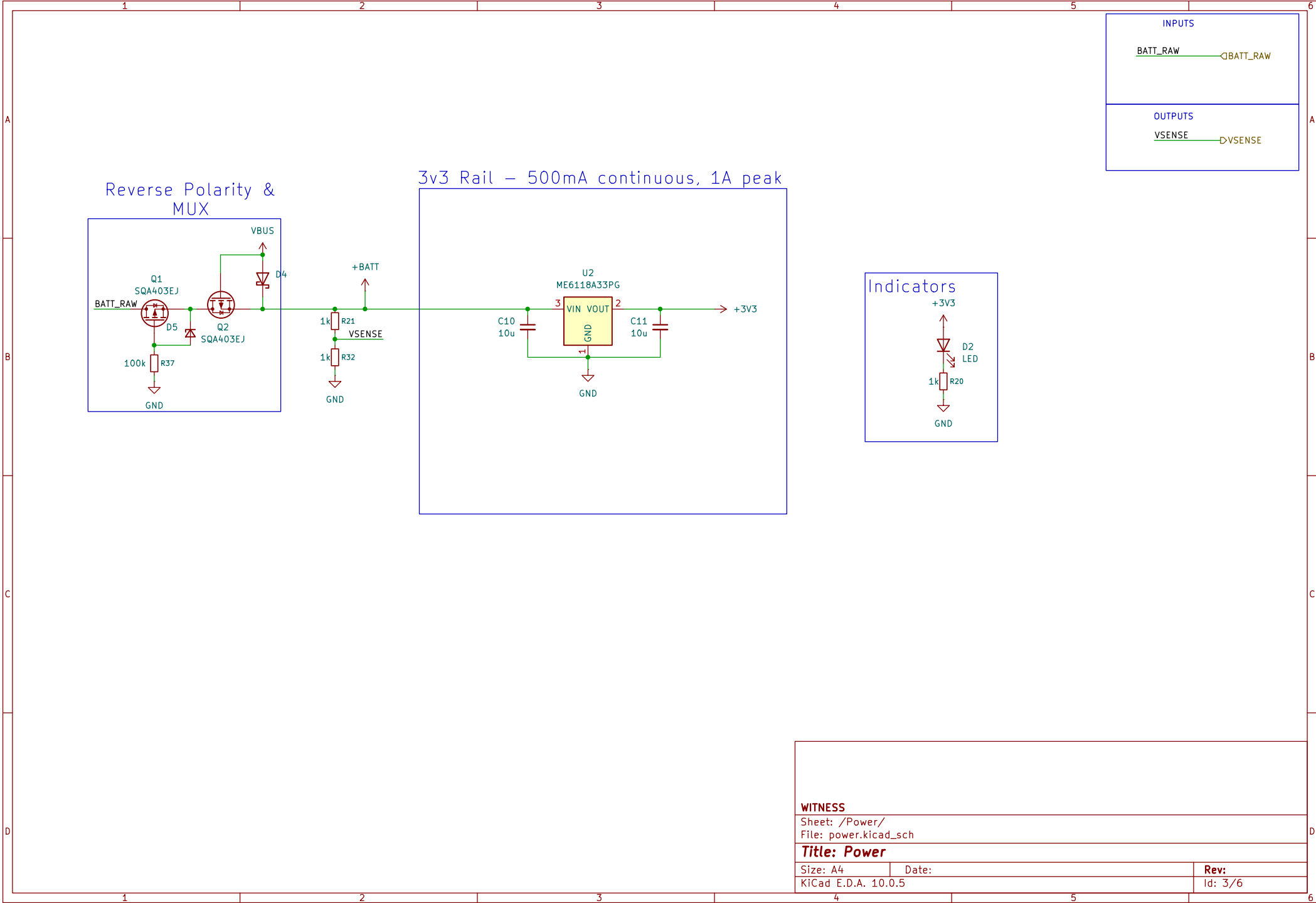
VDD_SPI power source ¹	VDD0P9_RTC via R _{PU}	VDD0P9_RTC	VDD0P9_RTC	VDD0P9_RTC
VDD0P9_RTC via R _{PU}	0	0	0	Ignored
VDD0P9_RTC via R _{PU}	1	1	1	Ignored
Flash Voltage Regulator	0	0	1	Ignored
Flash Voltage Regulator	1	1	1	Ignored

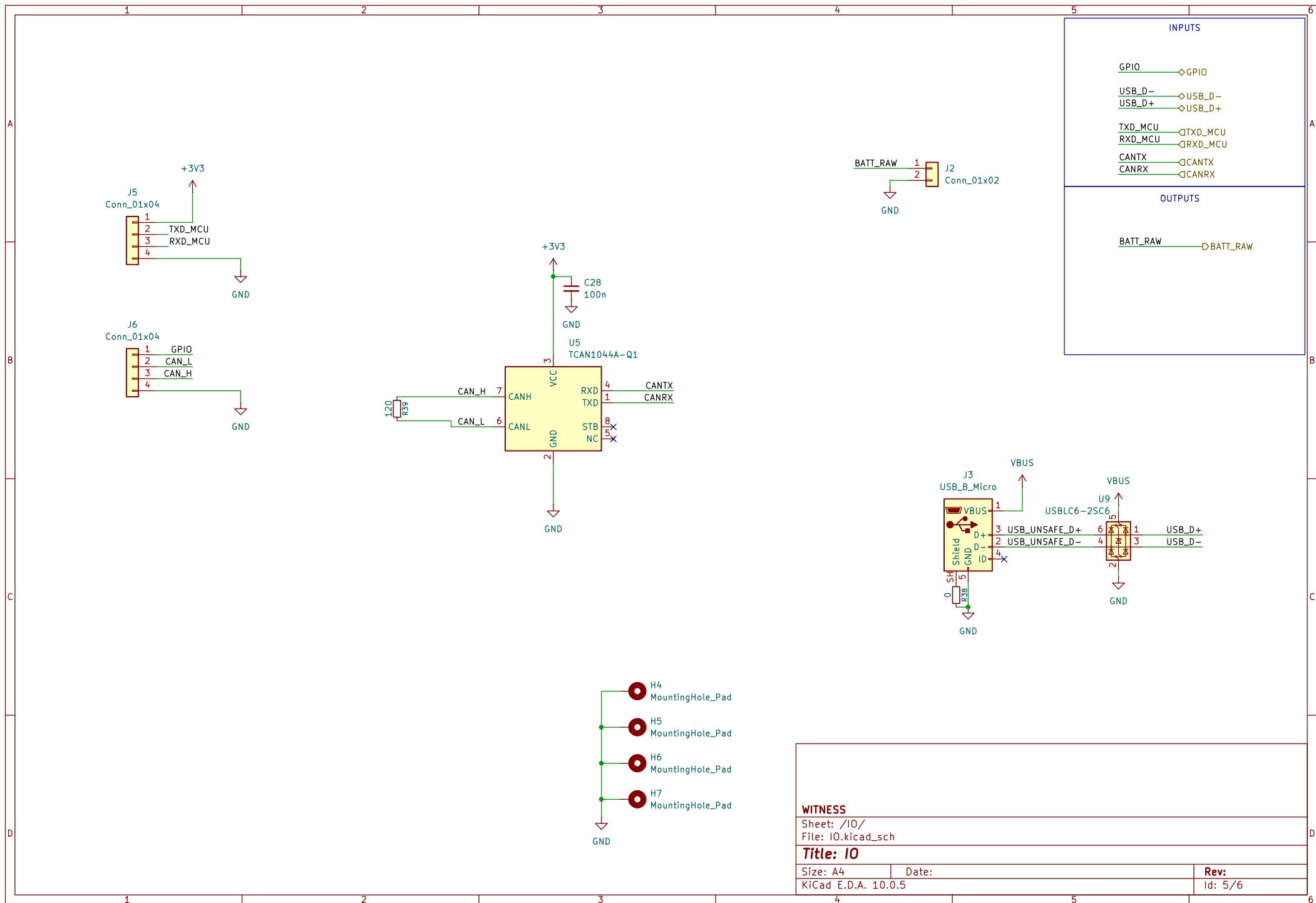
¹ Bold marks the default value and configuration.

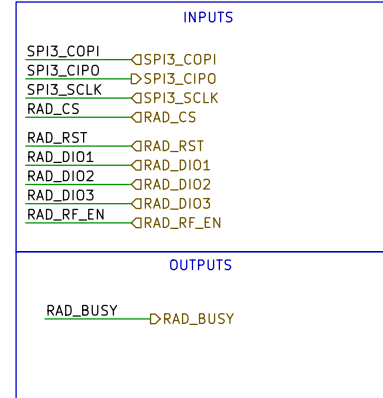
² See Section 3.3.5 Power Scheme.

Table 3-5. JTAG Signal Source Control

JTAG Signal Source	TFUSE_D15_PWD_JTAG	TFUSE_D16_USB_JTAG	TFUSE_STAMP_JTAG_SEL	GPIO3
	0	0	0	Ignored
USB Serial/JTAG Controller	0	0	1	1
	1	0	Ignored	Ignored
JTAG pins ²	0	0	1	0
	0	1	Ignored	Ignored
JTAG is disabled	1	1	Ignored	Ignored







KiCad E.D.A. 10.0.5